QP CODE: 25900201

MAHATHMA GANDHI UNIVERSITY, KOTTAYAM

SECOND SEMESTER MGU-UGP (HONOURS)

REGULAR EXAMINATION MARCH 2025

Second Semester

Discipline Specific Core Course: MG2DSCSAC100

MUSCULOSKELETAL SYSTEM AND BIOMECHANICS

(2024 ADMISSION ONWARDS)

SCHEME OF EVALUATION

Duration: 1.5 Hours **Maximum Marks: 50**

Part A (MCQ)

Answer all questions.

Each question carries 1 mark.

1. **What is the primary function of the muscles in the trunk?**

Answer: c) Protection and support for internal organs

2. **Which of the following nerves innervates the biceps brachii muscle?**

Answer: c) Musculocutaneous nerve

3. **Which plane of the body is most commonly used in imaging studies like MRIs and CT scans?**

Answer: a) Transverse plane

4. **At what muscle length is the maximum force typically generated according to the length tension relationship?**

Answer: c) At an optimal length where actin and myosin overlap maximally

5. **What is a motor unit?**

Answer: b) A group of muscle fibers activated by a single motor neuron.

6. **Which part of the skeletal muscle is responsible for contraction?**

Answer: b) Myofibrils

7. **The coronal plane divides the body into which two parts?**

Answer: c) Front and back halves

8. **Which of the following principles explains the relationship between the force applied to an object and its resulting motion?**

Answer: c) Newton’s second law of motion

9. **What function of the skeletal system is observed when an anatomical model shows the rib cage surrounding the heart and lungs?**

Answer: b) Protection of internal organs

10. **What is the main function of smooth muscle tissue?**

Answer: c) Regulation of internal organ movements

Part B (Short answer questions)

Answer 10 questions.

Each question carries 2 marks.

11. **Assess why the gluteus maximus is essential for sports performance.**
    The gluteus maximus provides hip extension, power, and stability, making it vital for sprinting, jumping, and explosive movements in sports.
12. **Describe the function of the sternocleidomastoid muscle.**
    The sternocleidomastoid rotates and flexes the neck, allowing head movement and stabilization.
13. **Differentiate between isometric and isotonic contractions in terms of movement.**
    Isometric contraction: No visible movement, muscle length remains constant. o Isotonic contraction: Movement occurs as the muscle changes length (either shortening or lengthening).
14. **How do bones help in movement?**
    Bones act as levers and provide attachment sites for muscles, allowing movement through joint articulation.
15. **What are joints, and what is their function?**
    Joints are connections between bones that allow movement and provide structural support to the skeleton.
16. **Describe the length-tension relationship in muscles and relate it to optimal force production.**
    Muscles generate the most force at an optimal length, where actin and myosin overlap efficiently. For example, a sprinter's hamstring functions best at a mid-range length during push-off.
17. **Evaluate the role biomechanics plays in enhancing athletic performance.**
    By optimizing technique, reducing energy waste, and improving movement efficiency, biomechanics enhances athletic performance through personalized training and injury prevention.
18. **Generate a biomechanics-based strength training program for improving muscular power.**
    Exercises: Plyometrics (box jumps), Olympic lifts (power cleans), and resistance training (squat jumps). o Focus: Enhancing SSC efficiency, optimizing movement mechanics, and improving force application.
19. **Explain how muscle force contributes to movement in human motion.**
    Muscle force generates tension, creating movement by pulling on bones via tendons. This enables actions like jumping, running, and lifting.
20. **Examine the role of Type IIb muscle fibers in explosive movements and power-based sports.**
    Type IIb fibers generate maximum force and speed but fatigue quickly. They are crucial for short, explosive actions like sprinting, weightlifting, and jumping.
21. **Describe the function of the myelin sheath and its role in neural transmission**.
    The myelin sheath is a fatty layer that insulates axons, increasing the speed and efficiency of electrical signal transmission, ensuring rapid nerve impulses for quick movements.

22. **Assess the effect of joint angle on muscle force production and provide practical applications in strength training.**
Joint angles affect leverage and muscle length, influencing force production. For instance, in bench pressing, adjusting the elbow angle optimizes force generation and reduces injury risk.

Part C (Essay type questions)

Answer 4 questions.

Each question carries 5 marks.

23. Compare the role of voluntary and involuntary muscles in maintaining body functions and supporting athletic performance.

- Voluntary Muscles (Skeletal): Controlled movements in running, weightlifting.
- Involuntary Muscles (Cardiac, Smooth): Maintain heart rate, blood circulation, and digestion.
- Cardiac Muscle: Sustains endurance in marathon running by pumping blood efficiently.
- Smooth Muscle: Regulates oxygen supply during exercise via airway dilation.
- Integration: Both muscle types work together for optimal performance.

24. **Discuss the significance of isokinetic contractions in rehabilitation and athletic performance.**

- Isokinetic training maintains a constant contraction speed, reducing injury risk.
- Used in rehab for controlled strength gains (e.g., knee injury recovery).
- Enhances muscular balance by avoiding dominant side overuse.
- Improves neuromuscular control (e.g., ACL injury rehab).
- Limited in sports application but valuable for post-injury strengthening.

25. Define joints and explain the different types of joints in the human body with examples.

- Definition: Joints are structures where two or more bones meet, allowing movement and stability.
- Types of Joints: 1. Fibrous Joints (Immovable) – Skull sutures. 2. Cartilaginous Joints (Partially movable) – Intervertebral discs. 3. Synovial Joints (Freely movable) – Most common in sports.
    - Hinge Joint – Knee, elbow (flexion/extension).
    - Ball-and-Socket Joint – Shoulder, hip (multidirectional movement).
    - Pivot Joint – Neck (rotation).
    - Gliding Joint – Wrist, ankle (sliding motion).
    - Condyloid Joint – Knuckles (flexion, extension, circumduction).
    - Saddle Joint – Thumb (opposition movement).

26. Compare the force-length and force-velocity relationships and their impact on muscle performance in training.

- Force-Length Relationship: Muscle force varies with length; optimal length allows maximal force production.
- Force-Velocity Relationship: As contraction speed increases, force generation decreases.
- Training Application: Strength training focuses on optimal muscle length and controlled velocities.
- Sports Performance: Understanding these relationships enhances sprinting, jumping, and lifting techniques.

• Injury Reduction: Prevents excessive loading at improper muscle lengths or speeds.

27. **Evaluate why understanding neural control of muscles is essential for optimizing sports performance.**
    • Motor Learning: Understanding how neurons control movement improves training techniques.
    • Strength Gains: Neural adaptations precede muscle hypertrophy, leading to faster strength gains.
    • Precision & Coordination: Improved neuromuscular control enhances accuracy in sports.
    • Reaction & Reflex Training: Essential for speed-based sports like sprinting, football, and boxing.
    • Injury Prevention: Proper neural control prevents overuse injuries and muscle imbalances.
28. **Examine how joint angle influences strength output in different exercises.**
    • Torque and Leverage: Strength varies with joint angles due to mechanical advantage.
    • Strongest Angles: Mid-range joint angles optimize muscle force and moment arm leverage.
    • Weakest Angles: Extreme flexion/extension reduces force due to suboptimal tension and leverage.
    • Exercise-Specific: Bench press is hardest at the bottom due to reduced leverage, while squats are weakest in the deep position.
    • Training Implication: Strength training should target weak points to improve full-range performance.